

Atelier 4:

JsonWebToken

1. Installation de JWT:

```
● houdaifa@houdaifa:~/everything/node/ateliers/atelier4$ npm install jsonwebtoken
added 13 packages, and audited 89 packages in 2s

22 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
❖ houdaifa@houdaifa:~/everything/node/ateliers/atelier4$
```

2. Require:

```
const mysql = require('mysql')
const jwt = require('jsonwebtoken')
const db = mysql.createConnection({
  host: 'mysql-houdaifa-alwaysdata.net'
```

3. verifyToken middleware:

```
// JWT Secret Key
const JWT_SECRET = 'secret'

// Verify Token Middleware
const verifyToken = (req, res, next) => {
  const token = req.headers['authorization']?.split(' ')[1]

  if (!token) {
    return res.json(error("No token provided"))
  }

  jwt.verify(token, JWT_SECRET, (err, decoded) => {
    if (err) {
      return res.json(error("Invalid or expired token"))
    }
    req.user = decoded
    next()
  })
}
```

4. login route:

```
// Login Route
app.post(config.rootAPI + 'login', (req, res) => {
  const { id, username, password } = req.body

  // Hardcoded user verification
  if (id === 1 && username === 'fullstack' && password === '123456') {
    const token = jwt.sign(
      { id, username },
      JWT_SECRET,
      { expiresIn: '1h' }
    )
    return res.json(success({ token }))
  }

  res.json(error("Invalid credentials"))
})
```

5. implementation de middleware:

```
app.use(config.rootAPI+'members', verifyToken, membersRouter)
//app.listen(config.port, () => console.log('Started on port '+config.port))
```

6. Tests:

a. Get avant login (sans JWT):

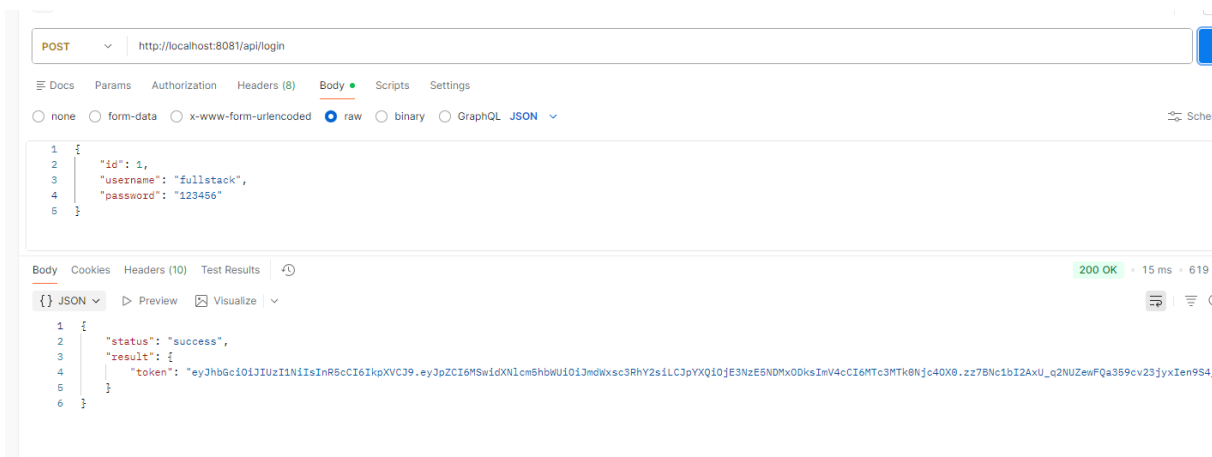
The screenshot shows a REST client interface with the following details:

- Method:** GET
- URL:** http://localhost:8081/api/members/
- Params:** Query Params table is empty.
- Body:** JSON format showing an error response.

Key	Value
-----	-------

```
{
  "status": "error",
  "result": "No token provided"
}
```

b. Login, recuperation de JWT:



POST http://localhost:8081/api/login

Body

```
1 {
2   "id": 1,
3   "username": "fullstack",
4   "password": "123456"
5 }
```

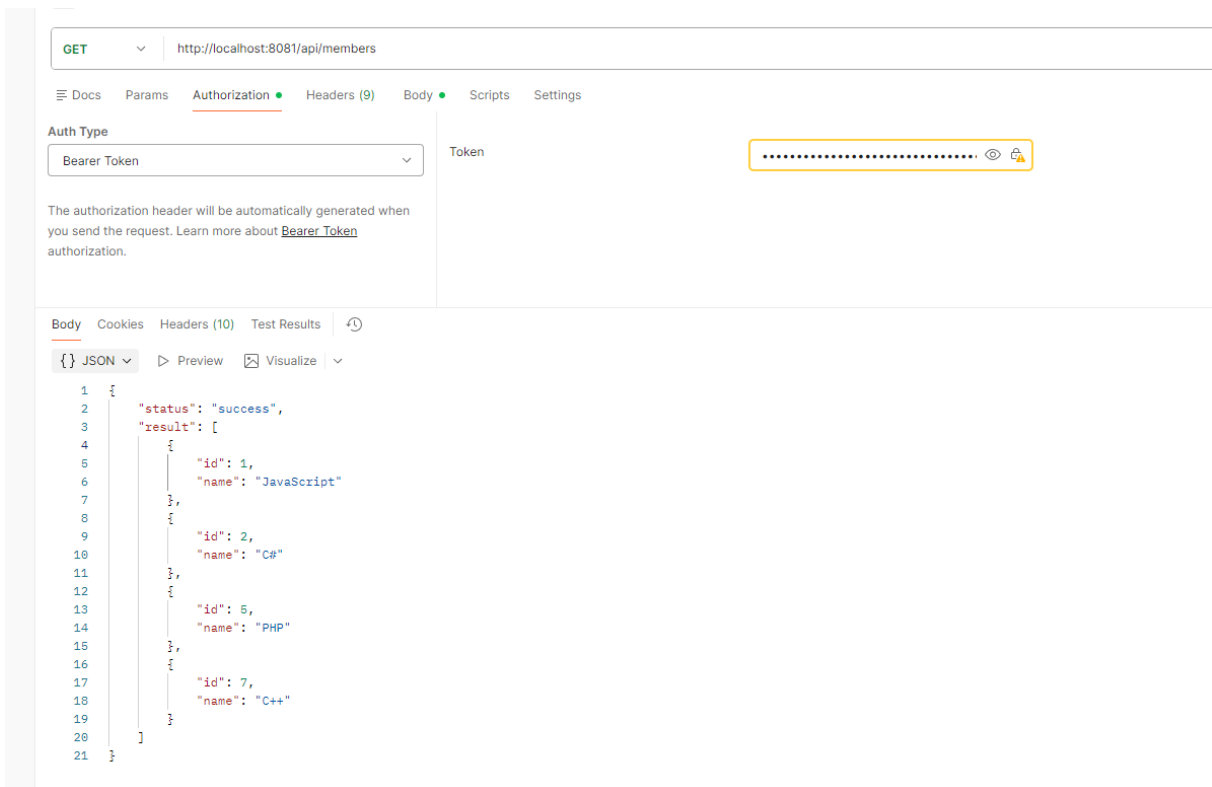
Body Cookies Headers (10) Test Results

JSON Preview Visualize

```
1 {
2   "status": "success",
3   "result": {
4     "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6MSwidXNlcm5hbWUiOiJmdWxsc3R5ZS1lC3pYXQ1bG9zE5NDMxODksImV4cCI6MTc3MTk4Njc4OXB8.zz7BNc1bI2AxU_q2NUZemFQa369cv23jyxIen9S4"
5   }
6 }
```

200 OK · 15 ms · 619

c. Get avec JWT:



GET http://localhost:8081/api/members

Auth Type: Bearer Token

Token: [REDACTED]

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body Cookies Headers (10) Test Results

JSON Preview Visualize

```
1 {
2   "status": "success",
3   "result": [
4     {
5       "id": 1,
6       "name": "JavaScript"
7     },
8     {
9       "id": 2,
10      "name": "C#"
11     },
12     {
13       "id": 5,
14       "name": "PHP"
15     },
16     {
17       "id": 7,
18       "name": "C++"
19     }
20   ]
21 }
```